

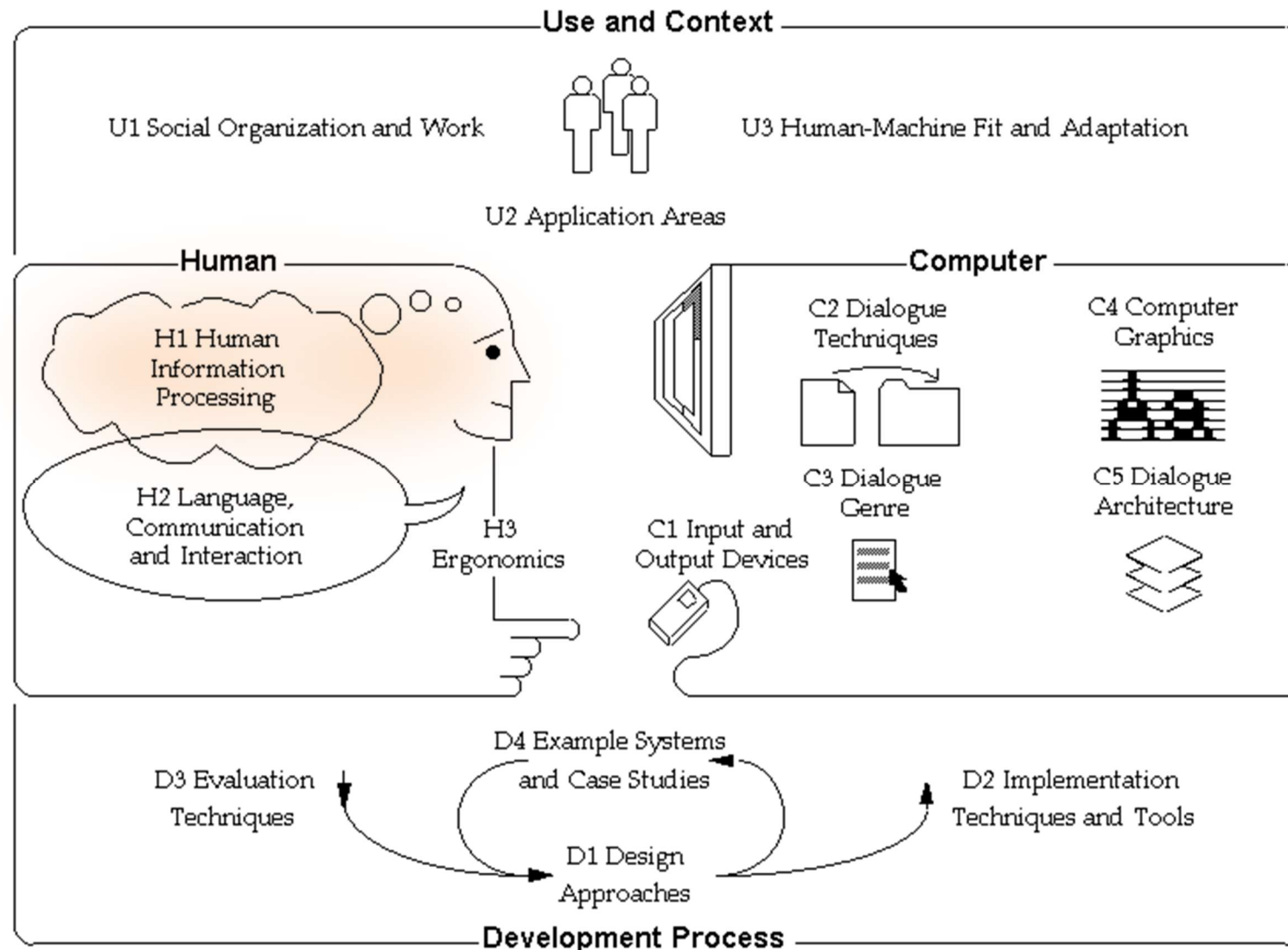
Human-Computer Interaction

Johan F. Hoorn and Jeff TANG

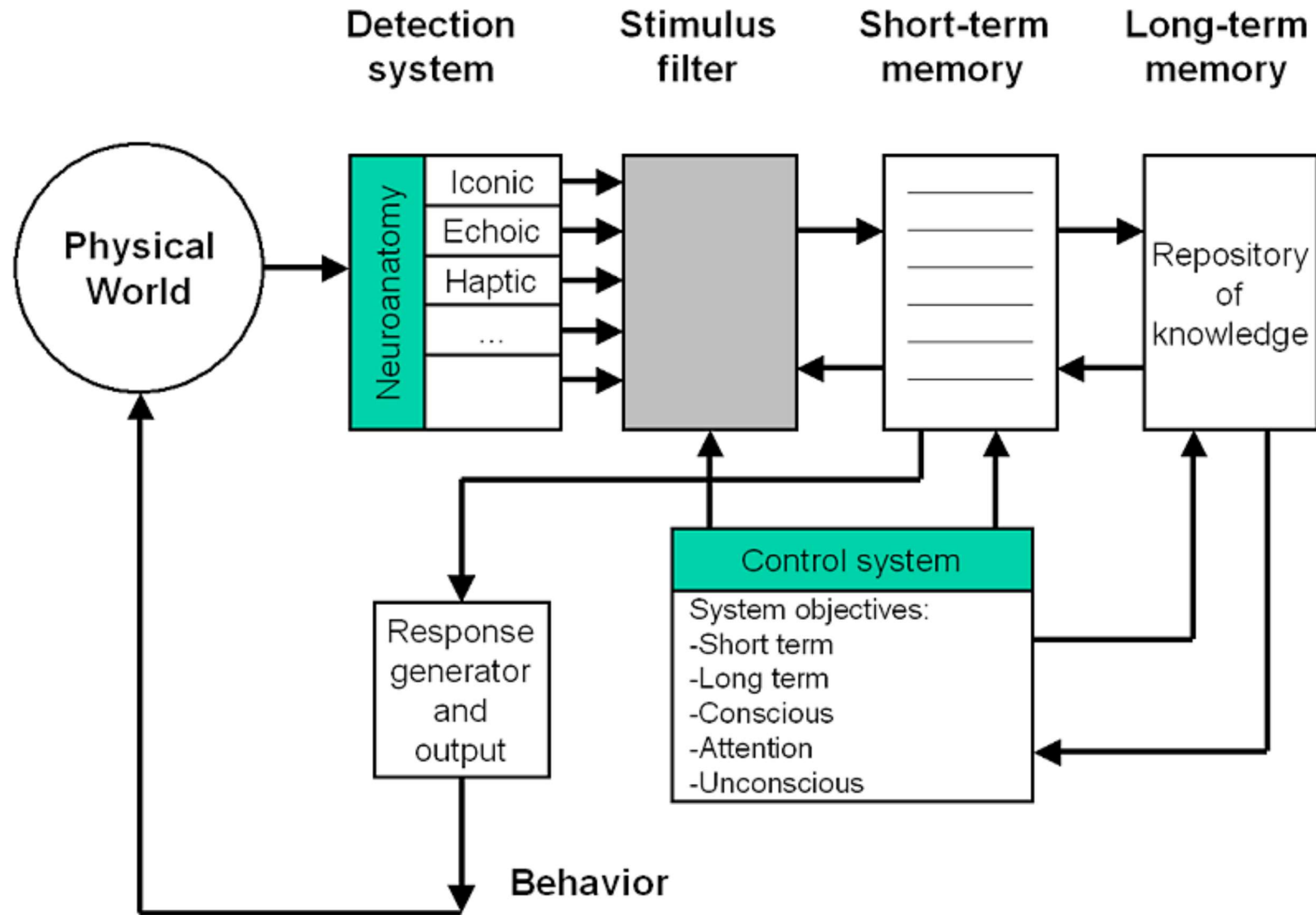
Human Information Processing, Part 2



Overview



The Human Information Processor



Problem representation

How you represent a problem (e.g., text or image) influences its solution

Text works better
to explain
abstractions:
'Communism'
'Democracy'

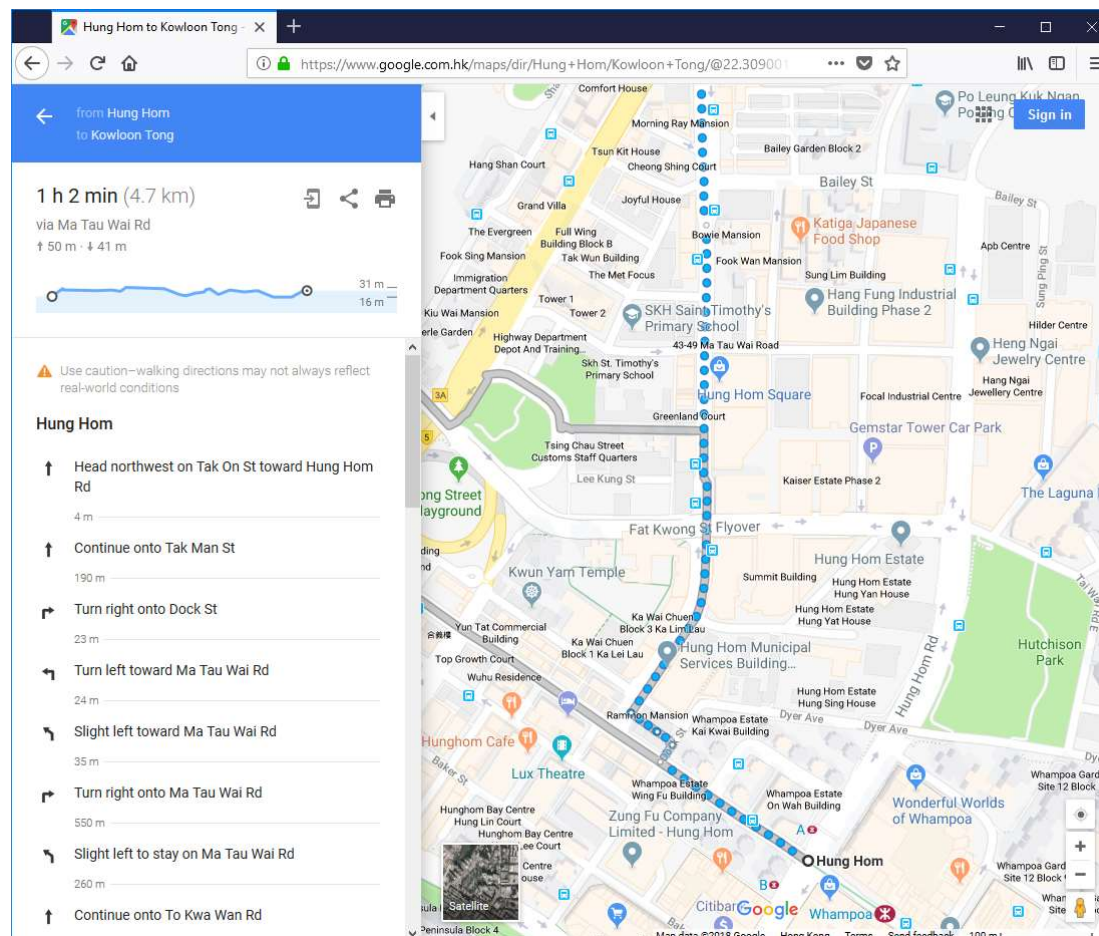


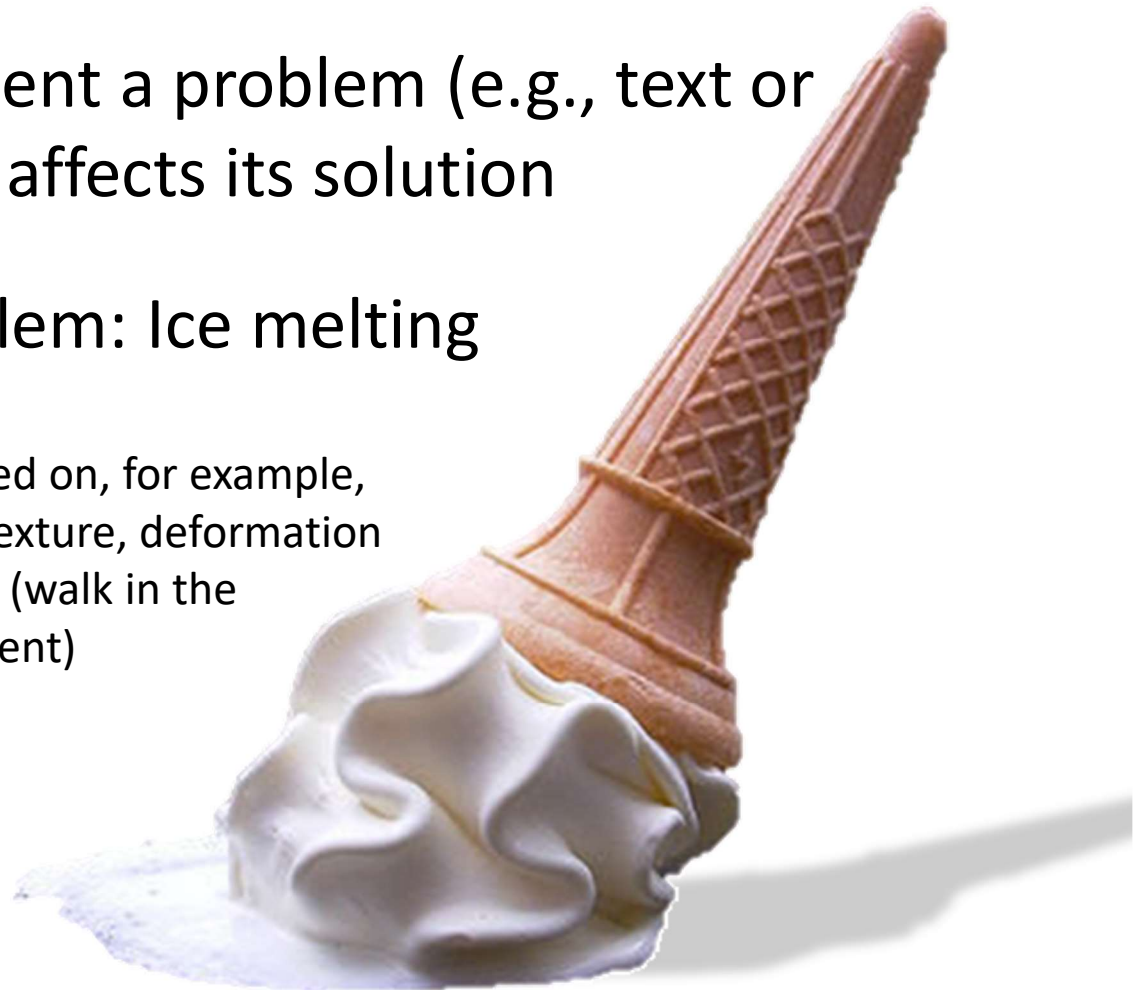
Image works
better for spatial
orientation and
localization. Where
is what?

Problem representation

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting

Solutions become focused on, for example, color changes, shapes, texture, deformation of solids, sun, and stains (walk in the shadow, napkins, detergent)

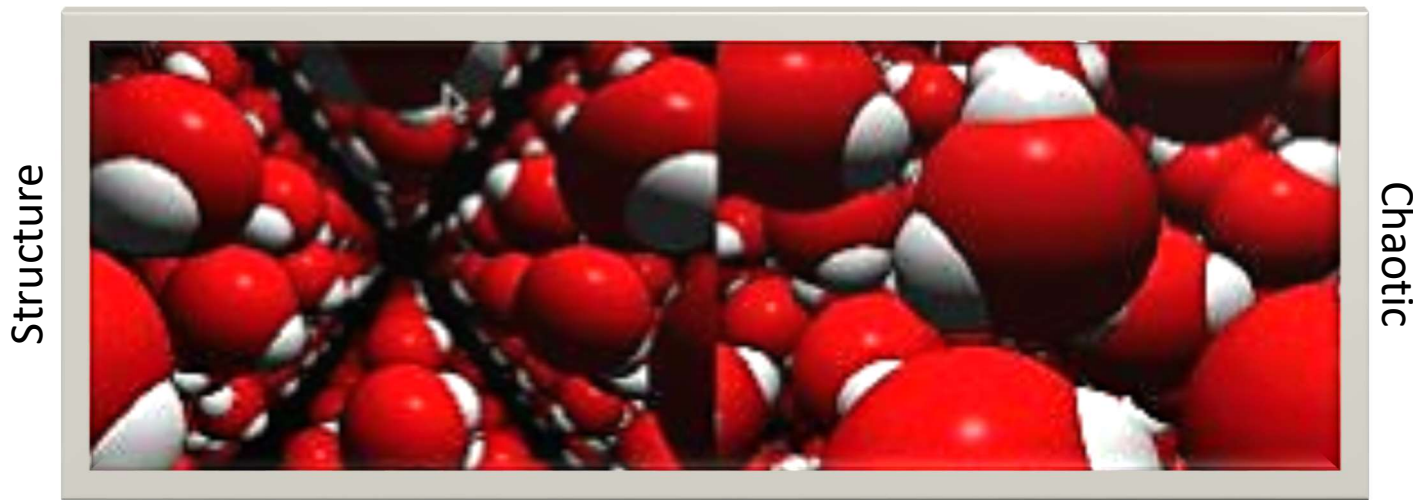


Surface level
Phenomenology

Problem representation

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting



Solutions directed to fixing loss of structure without creating gritty ice crystals and managing processes and mechanisms of, for instance, atoms and ions (e.g., add emulsifiers, more sodium chloride)

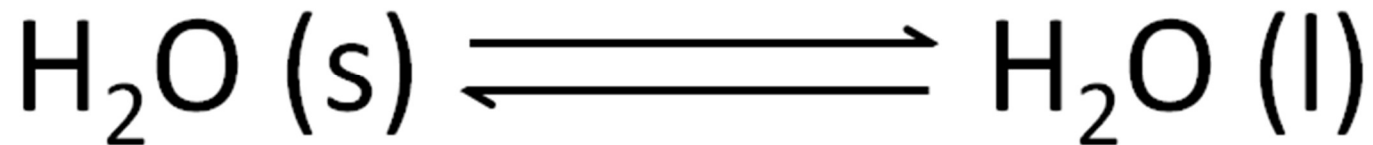
Structural level

Problem representation

3

How you represent a problem (e.g., text or image) affects its solution

Problem: Ice melting



Solutions derived from logics, reasoning, abstraction.

Protein BsIA binds with N_2 and O_2 (air), H_2O , and $-\text{COOH}$ (component of fat)

Symbolic level

Global warming solved: Scientists find ingredient that stops ice cream melting





VIDEO GAMES

MOVIES & TV

COMICS & COSPLAY

TABLETOP

SCIENCE & TECH

THE ESCAPIST PORTAL > THE ESCAPIST FORUMS > THE NEWS ROOM

Global Warming Solved: Scientists Find Ingredient That Stops Ice Cream Melting

THREAD INDEXSEARCH FORUMSCONTACT MODERATORSPOSTING GUIDELINES

PatrickJS

Muckraker





Posts: 260
Joined: 8 Jun 2015

1 POSTED: 1 September 2015 6:43 am

Global Warming Solved: Scientists Find Ingredient That Stops Ice Cream Melting

British scientists have isolated an ingredient that can keep ice cream from melting in the summer heat.

As the [U.N. convenes a closed-door meeting on climate change](#) this week, British scientists are one step ahead of the game on averting the worst kind of heat-related catastrophe: next summer, [your ice cream cone won't melt](#).

[Food researchers](#), from the Universities of Dundee and Edinburgh, have isolated a naturally occurring protein, BslA, that binds the air, fat and water in your triple scoop. In the end, this severely limits how quickly your frozen treat is going to melt.

The protein also keeps the ice cream frozen for longer, preventing gritty ice crystals from forming when you've kept the stuff in your freezer too long.

If you are interested in more nutritious food science news, you might like to [sample some bacon-flavored seaweed](#).

Cait MacPhee, a professor of [ice cream science](#) at Edinburgh and leader of the study, told the Telegraph, "we're excited by the potential this new ingredient has for improving ice cream, both for consumers and for manufacturers."

Personally, I'm not sure what the big deal is here. People have told me of this supposed 'melting,' but I've eaten three bowls of ice cream since I started writing this article and haven't noticed a thing.

Now, if only someone could invent ice cream that doesn't fall out of the cone just after you've paid for it...

Source: [Telegraph.co.uk](#)

[Permalink](#)



Image Source: McGill University

Solkard

2 POSTED: 1 September 2015 7:10 am

Problem representation

How you represent a problem affects its solution,
ALSO IN USER INTERFACES!

Symbolic level
emphasises
form transition from
solid to liquid matter
as logic reasoning



Visible laboratory level
emphasises e.g., color
changes, formation of
solids, and boiling

Molecular level
(Quicktime animation)
emphasises changes in structure and processes of,
for instance, atoms and ions

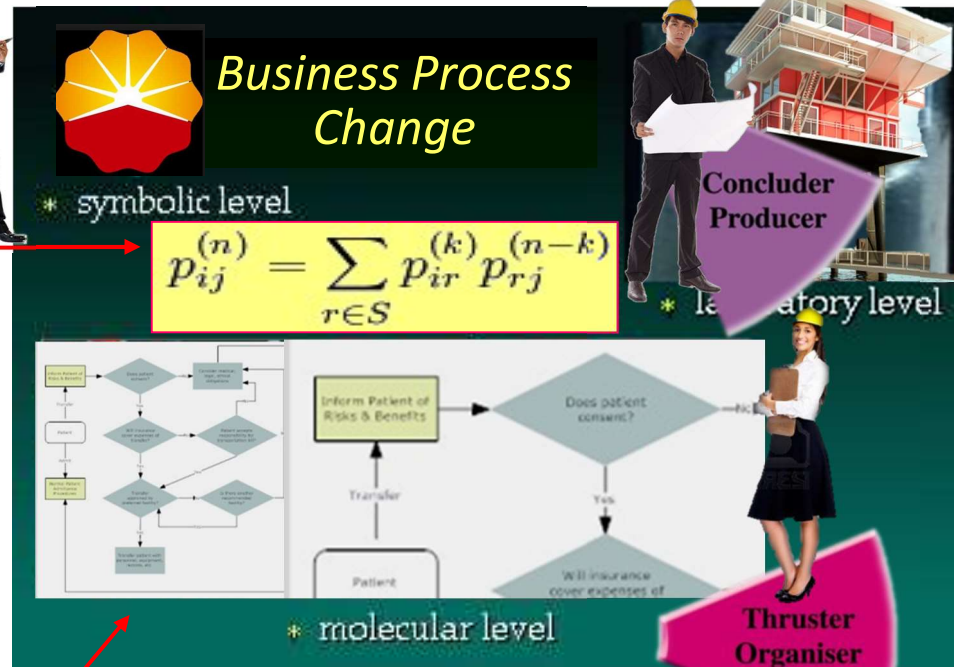
VisChem <http://www.uws.edu.au/science/Staff/rtasker/papers/images/interface.gif>

Change one level
and see how the
other levels change
as well. Insight!

Make multiple representations, each addressing another aspect

Problem: I want to solve this optimization problem

Symbolic level
emphasises
transition from
input to output
as logic reasoning



Visible laboratory level
emphasises,
for example,
type of workforce,
unforeseen mishaps,
work culture



Structural level
(Quicktime animation)

emphasises
changes in structure and processes of,
e.g., work units and their actions

So to improve the production, three sub-goals
took three different representations



As a designer, your clients will represent their problems in one specific way
Draw in more information than the problem representation that your client offers!

You copied everything from the Internet
We have a legal issue here

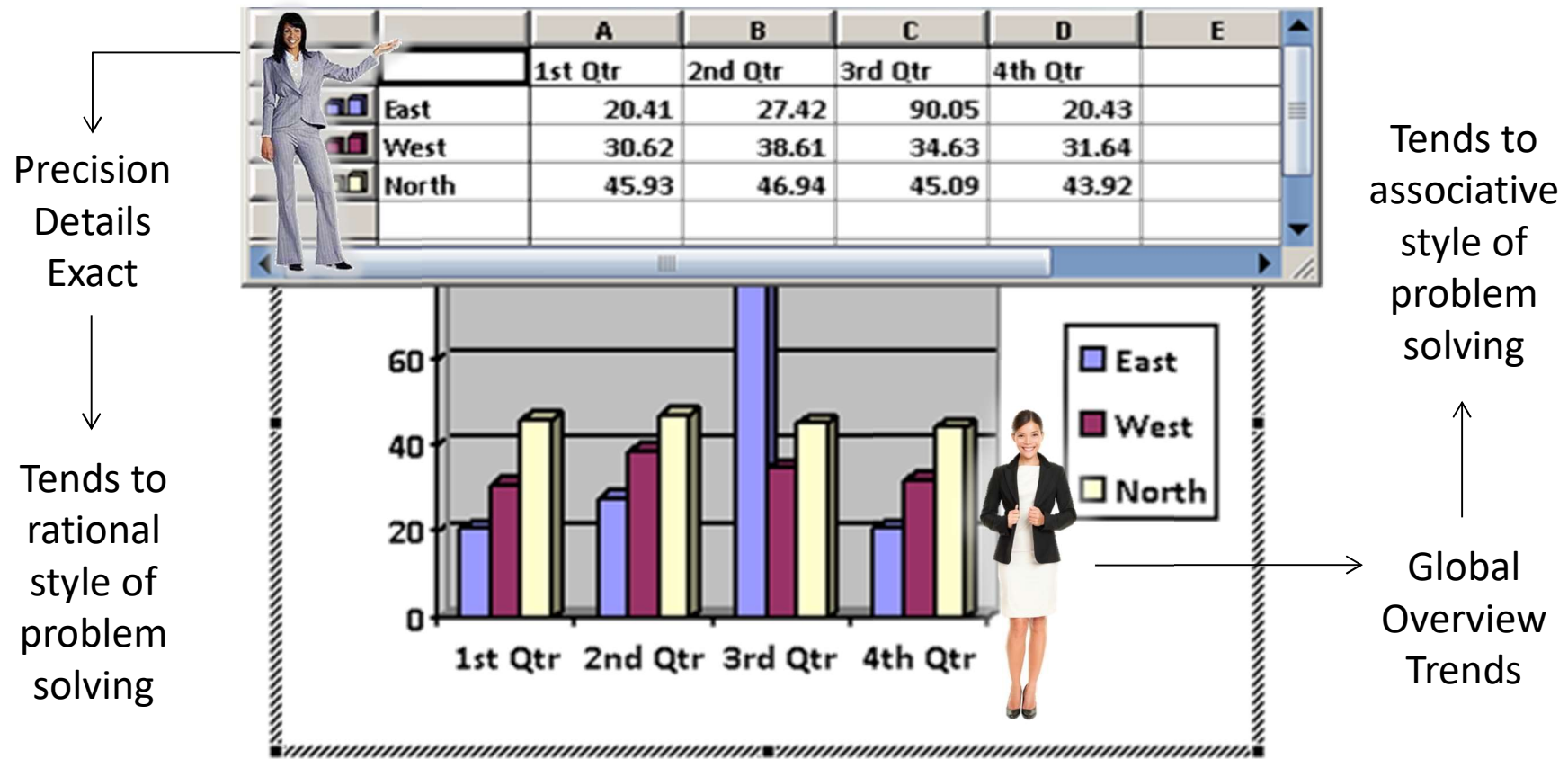
Let's try a new business model



Creative play is not just fun:
It is a form of problem finding and opportunity seeking!

In sum, how you represent a problem to your client (or your client to you), begs its solution

Each level of aggregating the data (cf. goal-decomposition) requires its own form of representation (cf. design)



Microsoft Word chart editor

Different mind sets solve problems differently

Who is your client (and who are you)?

The logic rationalist

~~COMP~~ Feeling happy in closed systems,
like a fish in a tank



The associative empiricist

~~SD~~ Feeling happy in open systems,
like a bird in a tree

Our students

The associative empiricist

Feeling happy in open systems

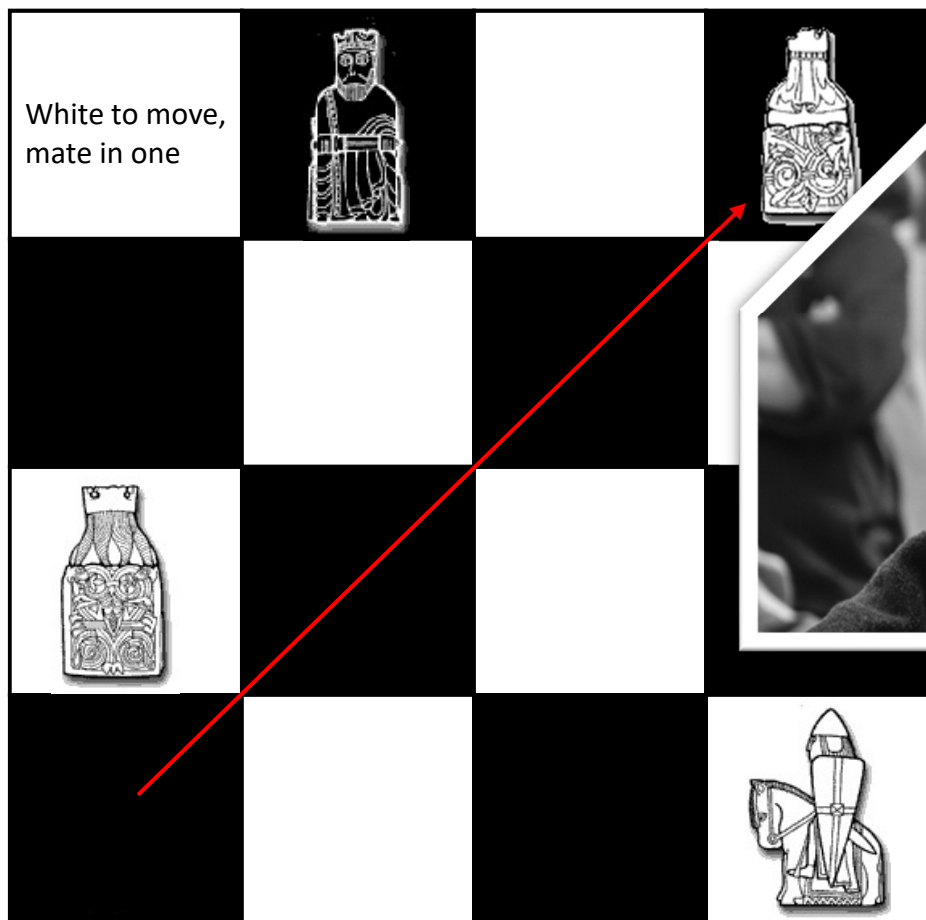


The logic rationalist

Feeling happy in closed systems



Structured decisions usually relate to well-defined problems and a more precise problem state



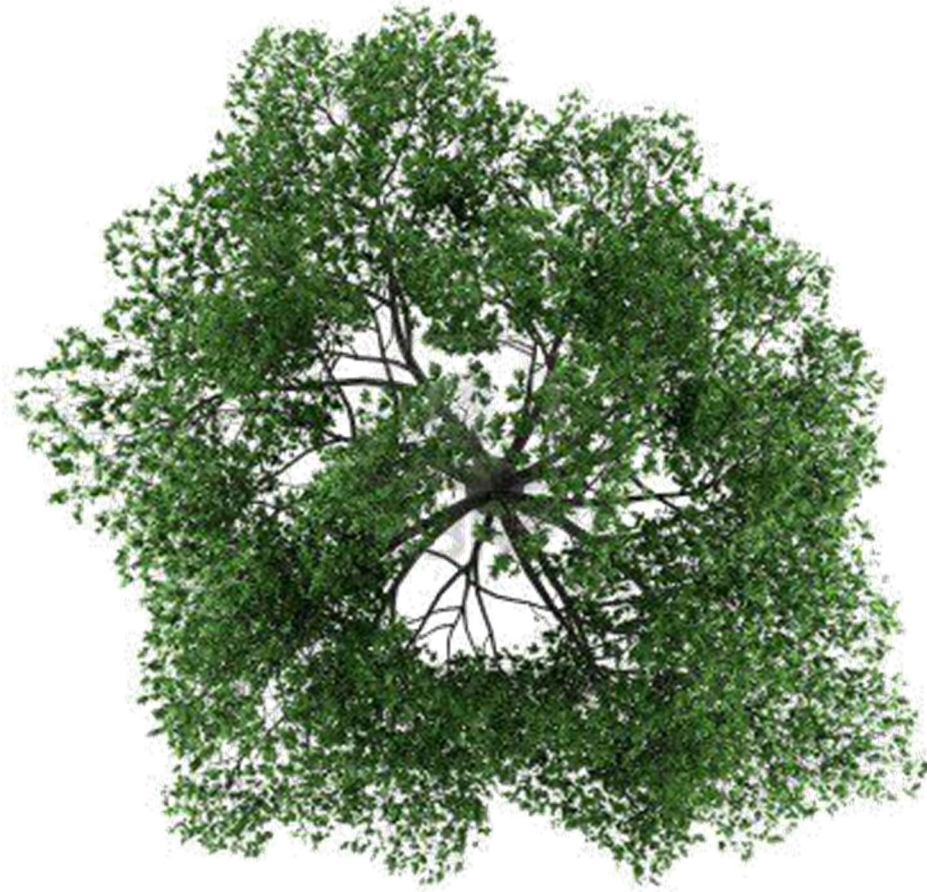
Unstructured decisions usually relate to ill-defined problems and a more vague problem state



What are you drawing?

Don't know.
I'm drawing.

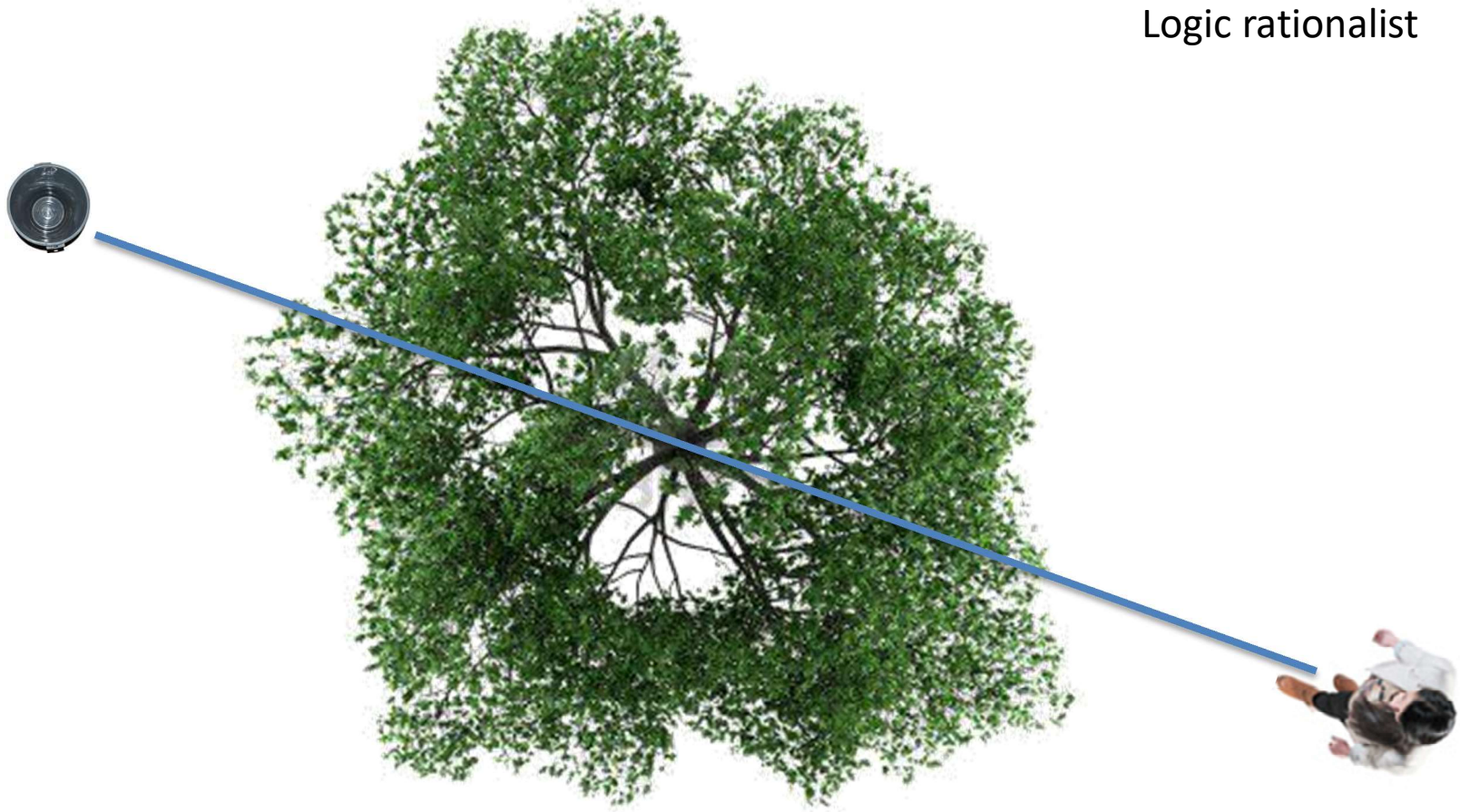
Tree - bin - me



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

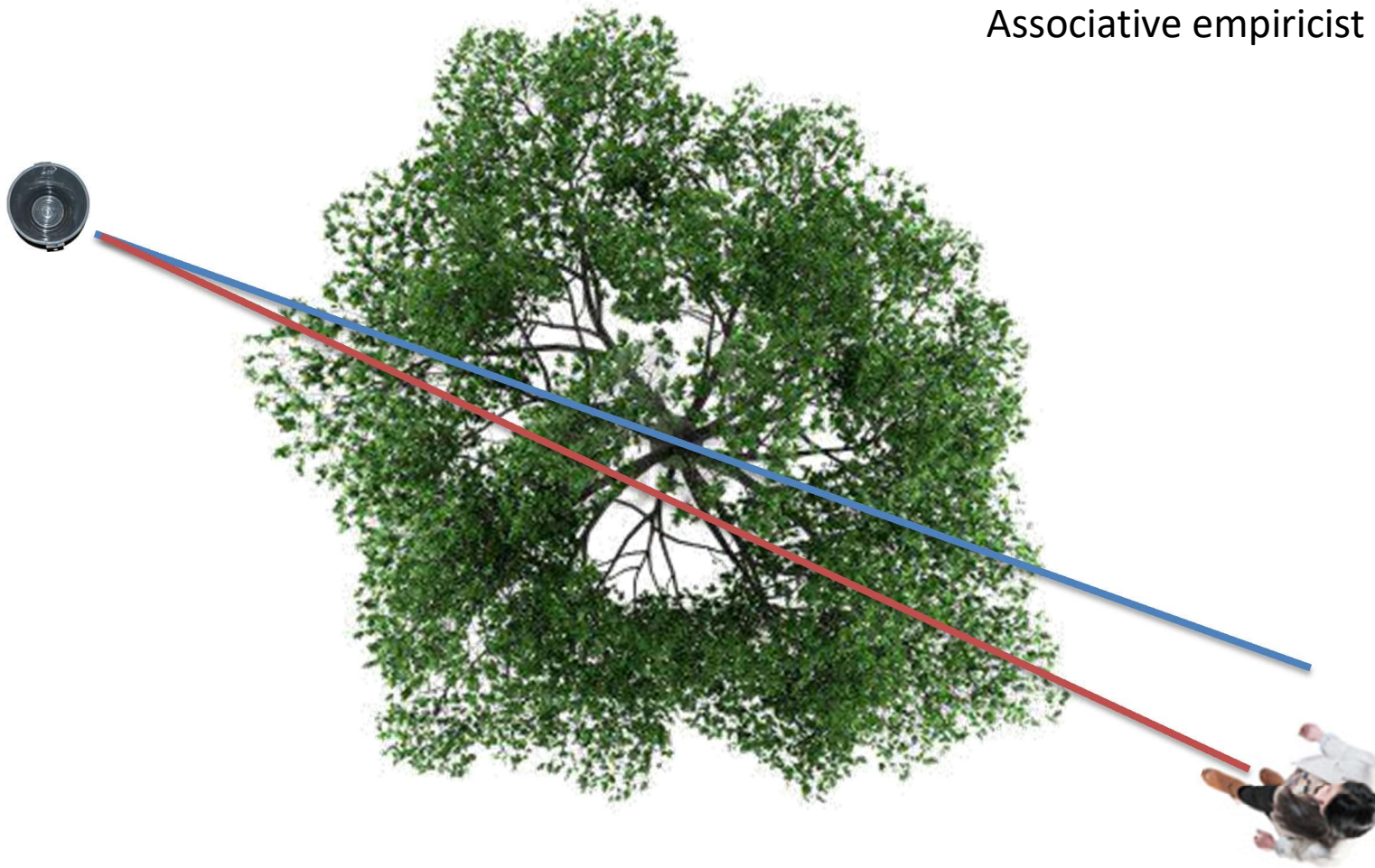
Logic rationalist



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

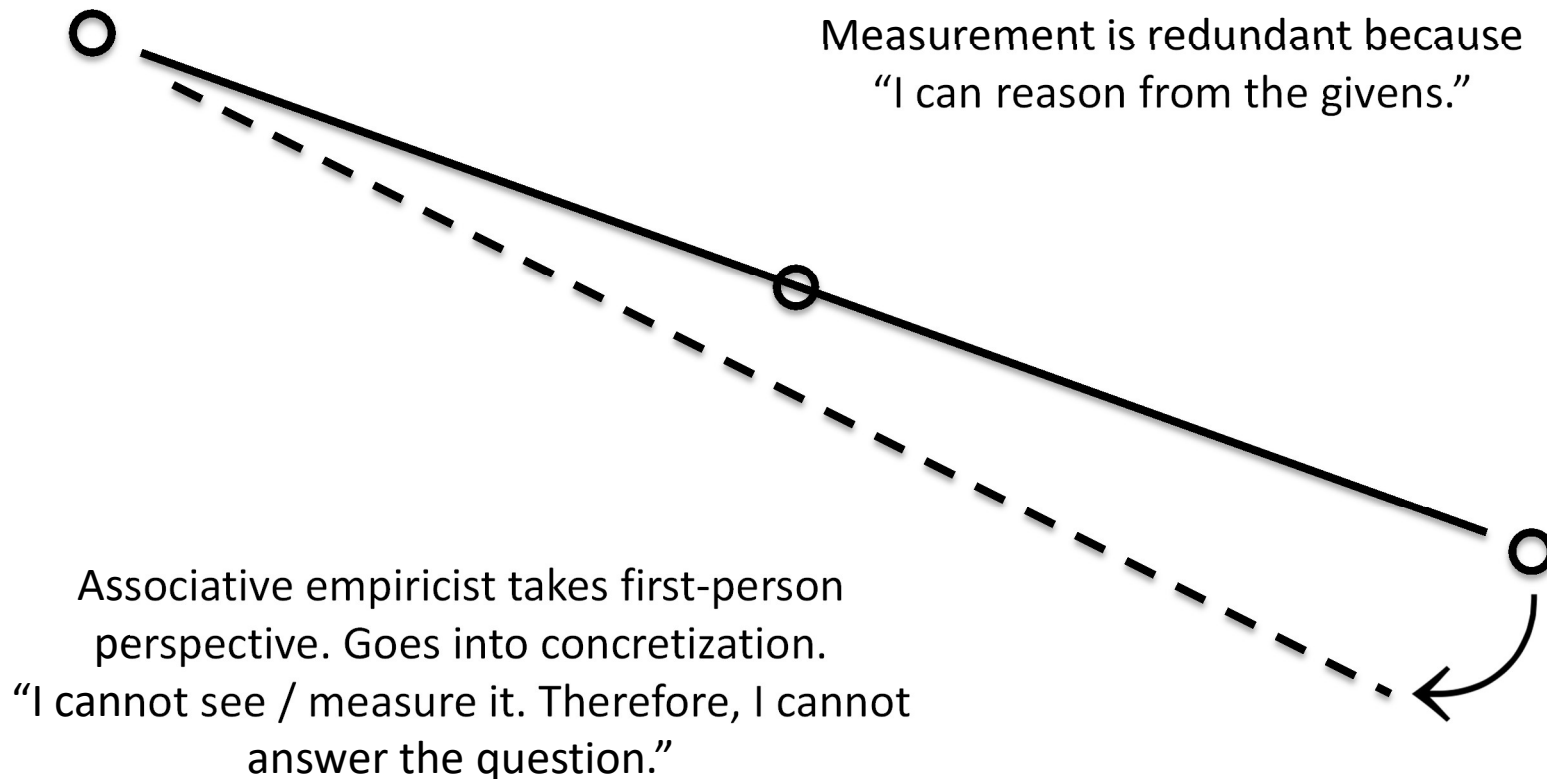
Associative empiricist



Position yourself such that the bin, the tree, and you form a straight line

Tree - bin - me

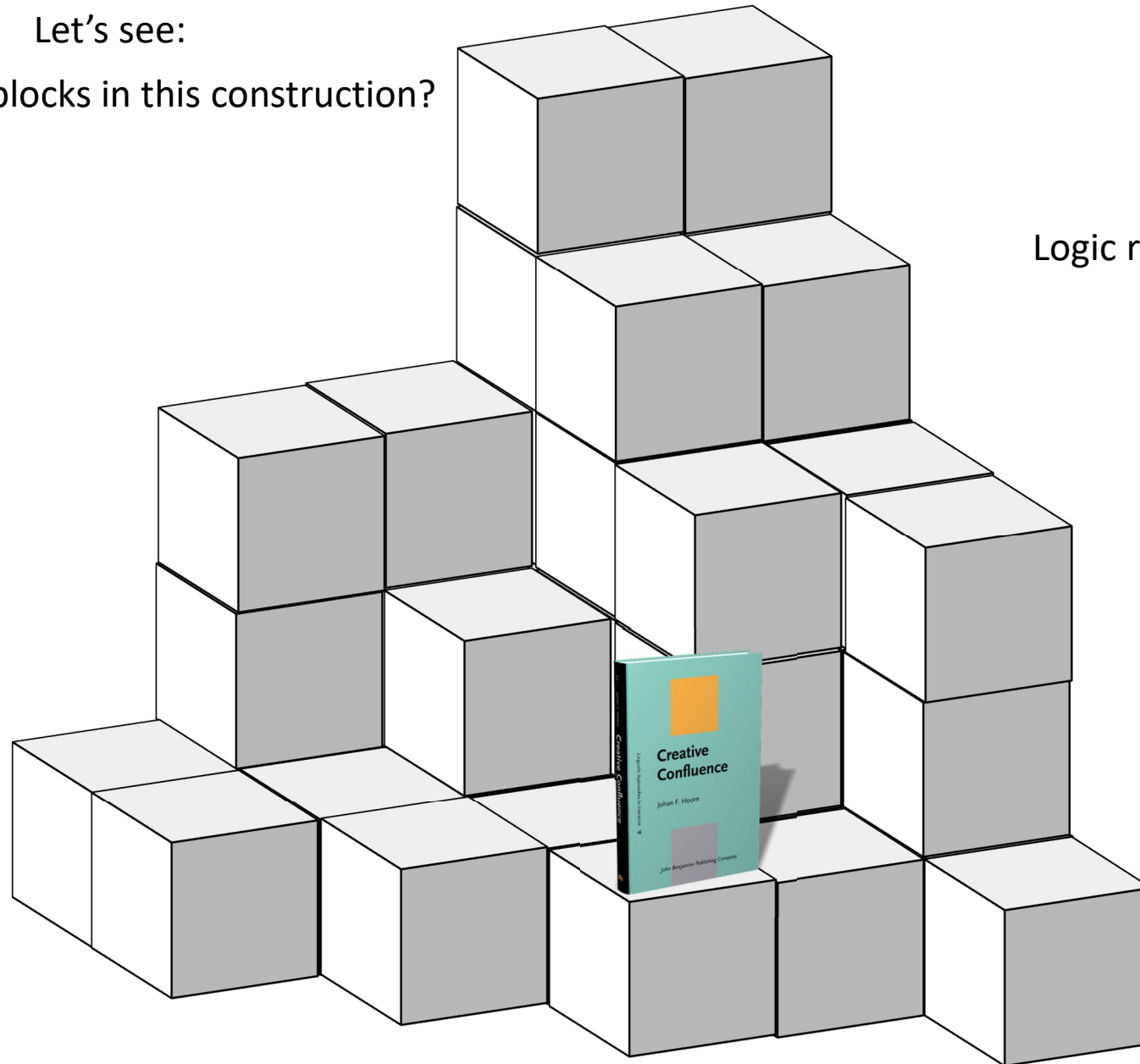
Logic rationalist takes person-independent overview.
Sees him/herself as a point in space.
Abstracts from the real world.
Measurement is redundant because
“I can reason from the givens.”



By now you probably think that
associative empiricists
are pretty stupid

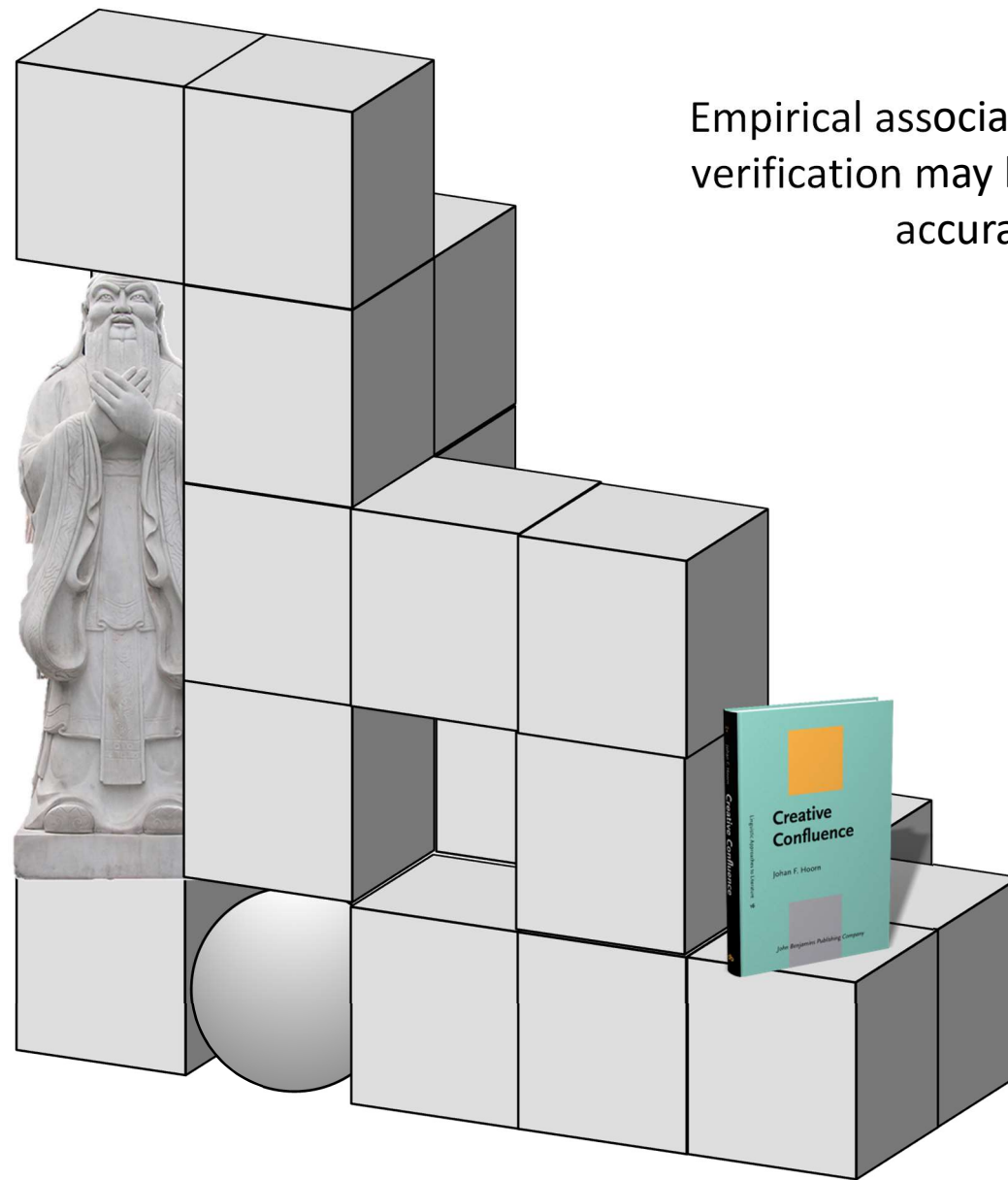


Let's see:
How many blocks in this construction?



Logic rationalism

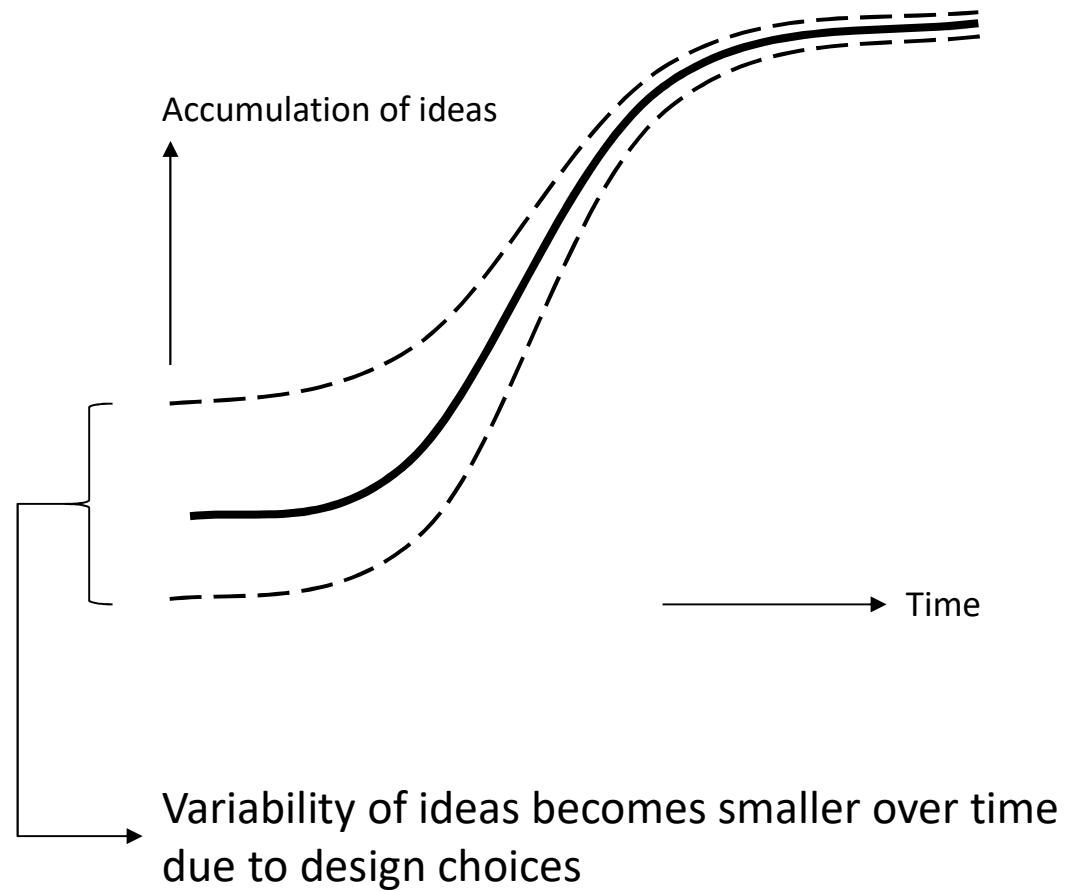
Hoorn, J. F. (2014). *Creative confluence* (p. 77). Philadelphia, PA: John Benjamins.



Empirical association and then
verification may lead to better
accuracy

Our students working together

Product life-cycle follows the Innovation S-curve



Break (5 minutes)

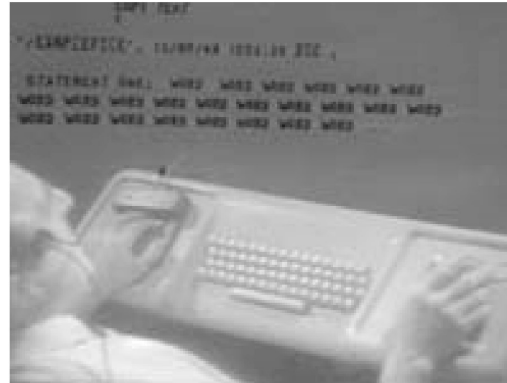


Output: response

❑ Motor

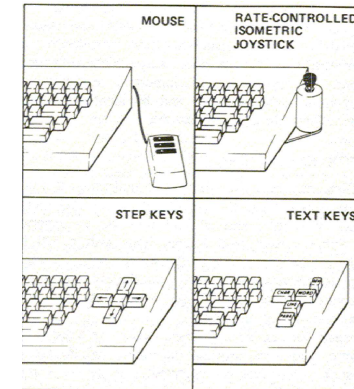
- Speed-accuracy trade-off
 - Fitts' law
 - Shannon's theory
- Performance evaluation
 - Index of Difficulty
 - Index of Performance

Problem solving

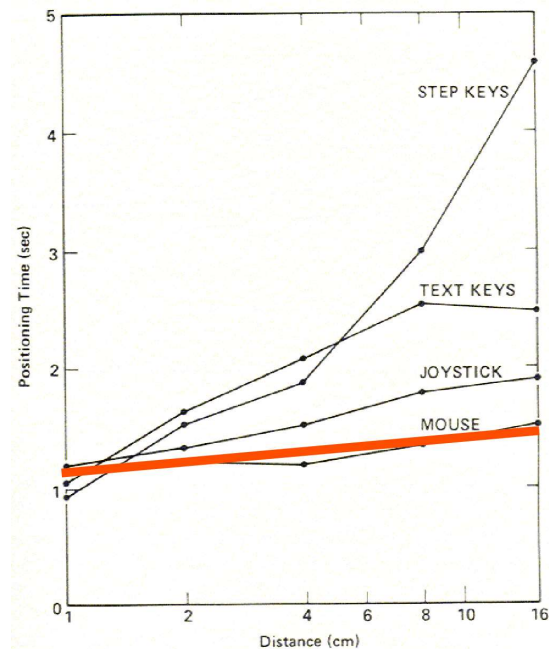


Engelbart: Which is fastest?

Multiple solutions



User testing (performance)



User testing (evaluation)

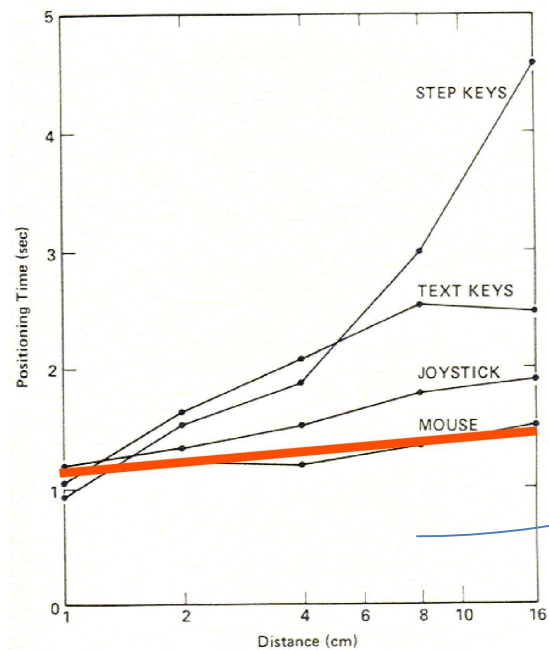




The 'Dislike' Button is finally coming

It looks like it's finally here, the Dislike button for Facebook. CEO Mark Zuckerberg has for many years said they wouldn't add a "dislike" button, but in a Question and Answer session Tuesday, Zuckerberg let the world know they will soon be adding one.

"I think people have asked about the dislike button for many years. Today is a special day because today is the day I can say we're working on it and shipping it," said Zuckerberg.

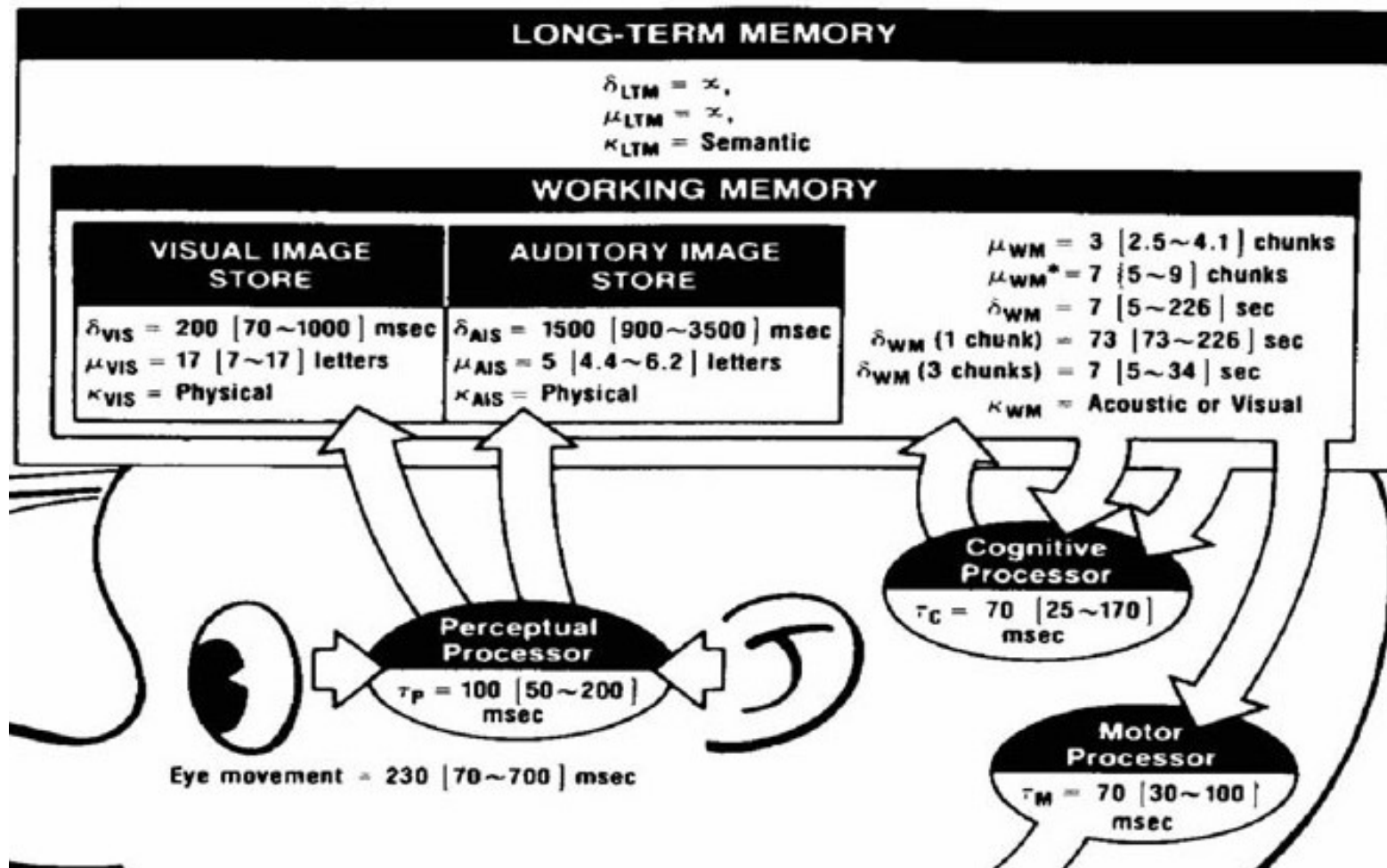


Attitude

Time scale of human behavior

10^7 (months) 10^6 (weeks) 10^5 (days)	Social	Social Behavior
10^4 (hours) 10^3 10^2 (minutes)	Rational	Adaptive Behavior
10^1 10^0 (seconds) 10^{-1}	Cognitive	Immediate Behavior
10^{-2} 10^{-3} (milliseconds) 10^{-4}	Biological	

Performance

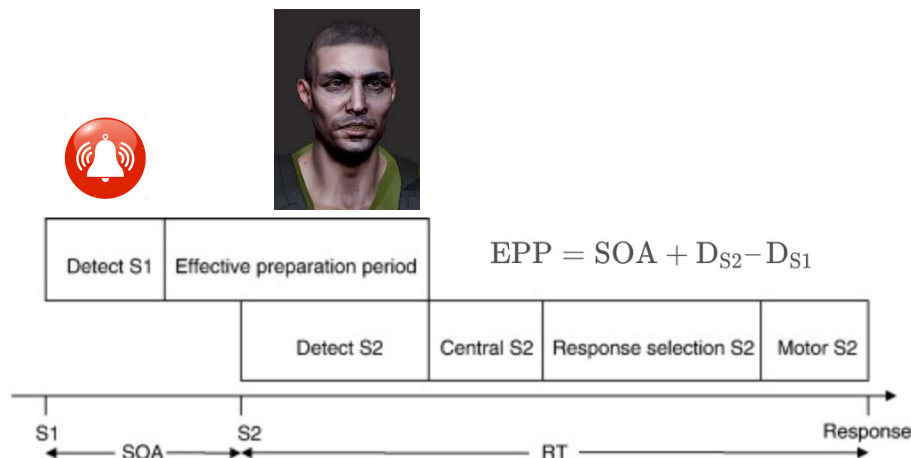
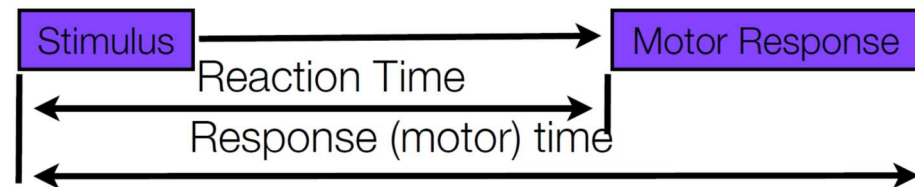


Output: response time

□ Motor

- Time taken to respond to stimulus (speed): reaction time + movement time
- Movement time is dependent on physical characteristics of subject: e.g. age, fitness level
- Reaction time is dependent on stimulus type:

- Visual: ~200 ms
- Auditory: ~1500 ms
- Pain: ~700 ms



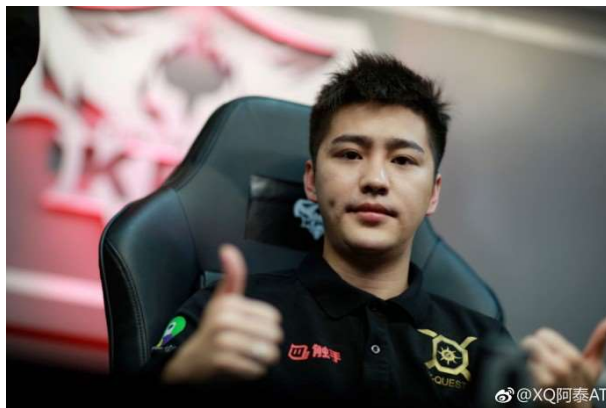
Remember trigger-happy
game research? Lecture 01:
Use and Context

Output: accuracy of response

❑ Motor

- Another measure of motor skill is accuracy
- There is a trade-off between speed and accuracy:
 - Faster is sloppier (inaccurate)
 - Slower is precise (accurate)
- However, whether speed of reaction will result in reduced accuracy depends on the task and the user

That is why you do training: fast + accurate



AT, professional gamer, *Honor of Kings*



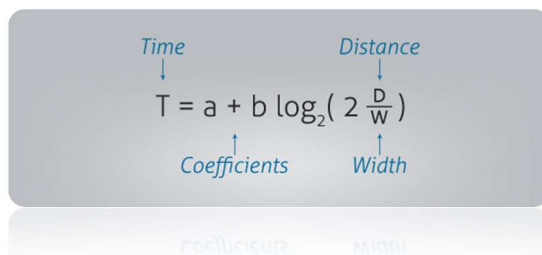
Fafa Yim, Logitech pro-team captain, *League of Legends*
Faster than AT with more target hits

Output: speed vs accuracy

❑ Motor

Fitts' Law: speed vs accuracy

- Established in 1954 by Prof. Paul Morris Fitts
- Lieutenant-colonel of the US Air Force
- Head of the US Air Force Human Engineering Division
- Later professor at Ohio State U and U of Michigan
- A model of human movement
- Predicts the time required to rapidly move from a starting position to a final target area
- Models the act of pointing
- In the real world (e.g., with a hand or finger, or even eye tracking)
- On computers (e.g., with a mouse or a joystick)



The diagram shows the Fitts' Law equation $T = a + b \log_2 \left(2 \frac{D}{W} \right)$ inside a light blue rounded rectangle. Labels with arrows point to parts of the equation: 'Time' points to 'T', 'Coefficients' points to 'a' and 'b', 'Distance' points to 'D', and 'Width' points to 'W'.

$$T = a + b \log_2 \left(2 \frac{D}{W} \right)$$

The wider an interface widget and the closer it is to the pointer, the easier to hit it

(Lecture 01: Use and Context)

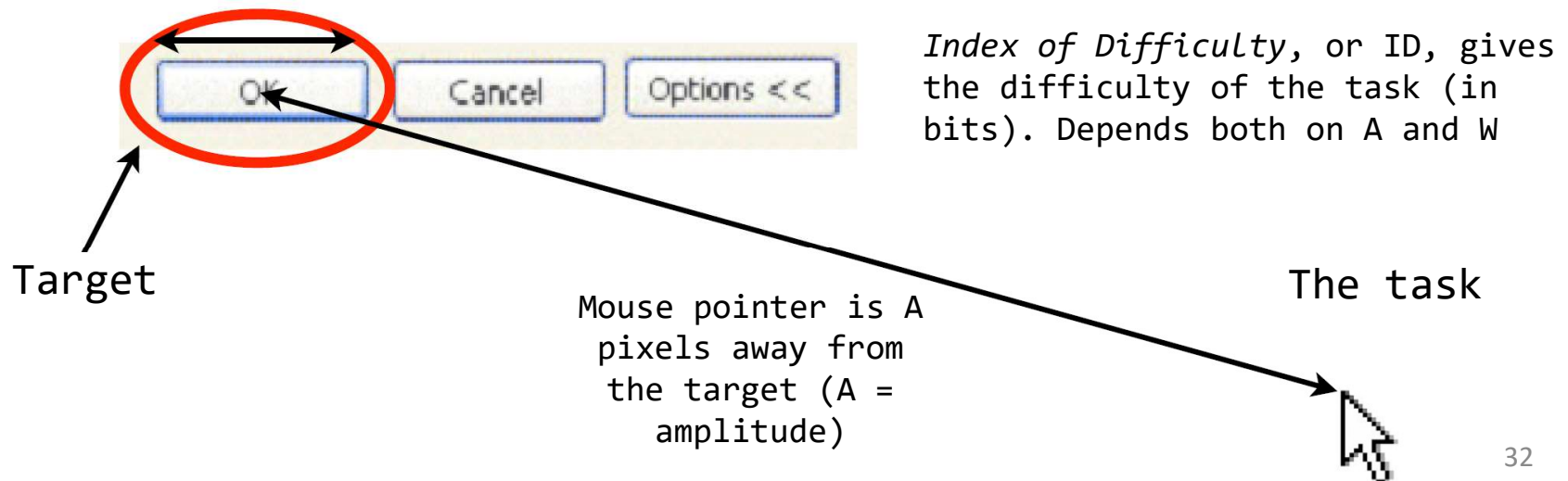
Output: speed vs accuracy

□ Motor

Fitts' Law: speed vs accuracy

- Model of human movement
- Based on Shannon's theory of information systems
- Human motor system modeled as channel used to transmit information
- Task has a certain difficulty (measured by number of bits)
- Channel has a certain bandwidth (measured by seconds/bit)

Target is W pixels wide (W = tolerance)



Output: speed vs accuracy

Index of Difficulty

- Movement to target requires more time when distance to target (A) increases
- Error Tolerance of target (W) decreases
That is, when target is farther away or of smaller size
- Movement to target requires less time when distance to target (A) decreases
- Error Tolerance of target (W) increases
That is, with a closer target or one of larger size
- Originally: $I_D = \log_2(2A/W)$
- Shannon-style reformulation:
- $I_D = \log_2(A/W + 1)$, or $I_D = \log_2(A/W + 0.5)$ (when $I_D < 3$ bits)

The diagram shows the equation $T = a + b \log_2(2 \frac{D}{W})$ inside a light blue rounded rectangle. Arrows point from labels to parts of the equation: 'Time' points to 'T', 'Coefficients' points to 'a' and 'b', 'Distance' points to 'D', and 'Width' points to 'W'.

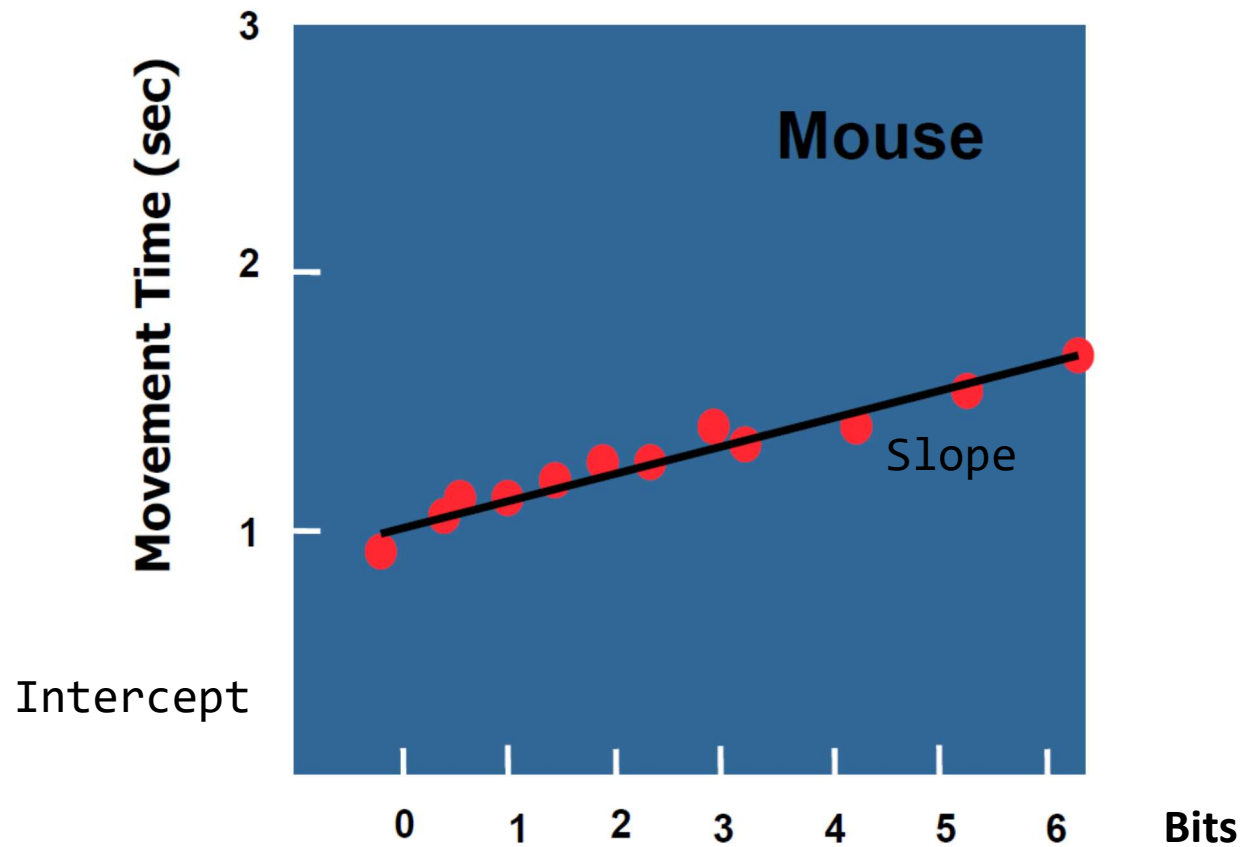
$$T = a + b \log_2(2 \frac{D}{W})$$

The wider an interface widget and the closer it is to the pointer, the easier to hit it

(Lecture 01: Use and Context)

Output: speed vs accuracy

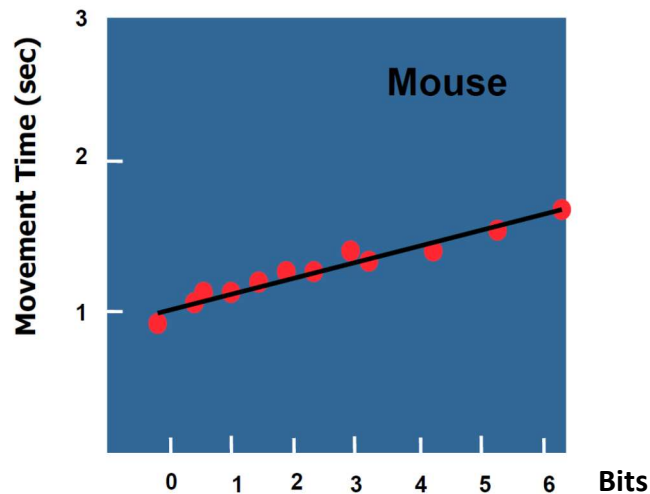
The channel is characteristic of the modality of the interaction



Output: speed vs accuracy

Putting everything together:

- Time T_{pos} needed to complete the task is linear with respect to ID
- $T_{pos} = a + b \cdot ID$
- a and b are empirically determined through experimental data
- a = y-intercept
- b = slope (secs/bit)
- T_{pos} = time taken (seconds)



Output: speed vs accuracy

Evaluating a device

Index of Difficulty depends on the task, not the device

The regression constants a and b determine how fast the task is done (or how fast the device processes the task)

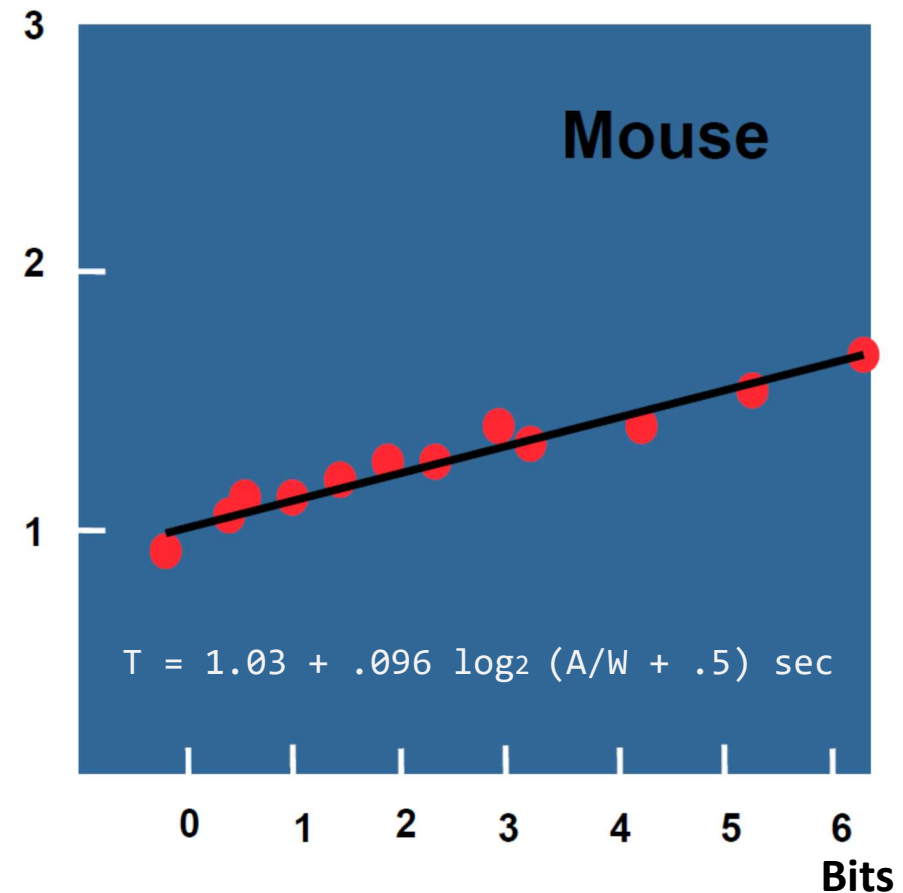
The *Index of Performance* IP (or *throughput* TP) defines how quickly the task can be finished

$$IP = 1/b$$

Aside: the mouse is very close to being an optimal device
(hard to do better than the mouse)

$$I_D = \log_2(\text{Dist}/\text{Size} + .5)$$

Time



Virtual Tutor Summary



Human Information Processing is different from computation because perception, memory, problem solving, and (motor) responses are less 'stable' but also more adaptive

All the visual, textual, sound, and touch input become an integrated whole and recede in memory, sometimes in a single cell (e.g., faces and places)

Memory is not fixed but changes over time (neurons change their connections). Memory is context-sensitive

Problem-solving skills may differ. For deterministic problems, you need reasoning (e.g., means-end analysis). For underdetermined problems, you need creative solutions

Virtual Tutor Summary



When you design a device, you need to reckon with speed-accuracy trade-offs. That is, trained users can be highly efficient, combining high speed with great accuracy (cf. pro-team gamers)

It also means that untrained users (novices, first-time users, older adults) may not be as advanced as you

User testing of speed and accuracy tells you how your system or device does in terms of user performance. Don't forget to ask for appreciation as well (like-dislike)

Fitts' law as derived from Shannon's Theory of Information Systems is your means to determine user performance of motor output

Exam Sample Item

Category “Knowledge”

When can you consider a problem to be ‘well-defined’?

A When goal decomposition allows to break down a complex problem into smaller ones by determining sub goals

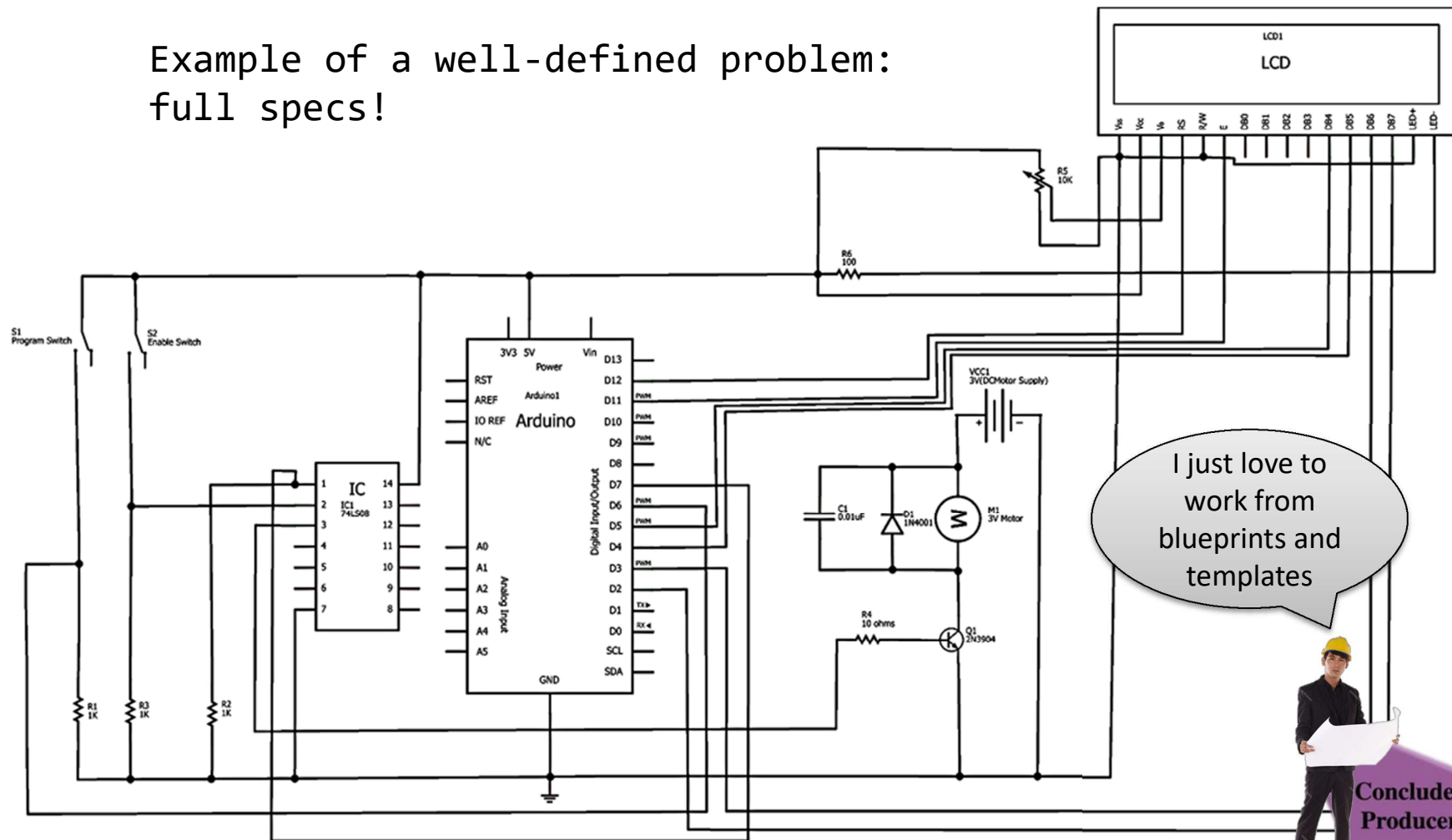
B When structured decisions are made in a problem state that has multiple representations



C With full specification of the provided data, actions, strategies, and goal states, which can be represented in a problem space

D When at the structural level, the focus lies on changes in structure and processes such as work units and actions

Example of a well-defined problem:
full specs!



I just love to
work from
blueprints and
templates



Concluder
Producer

In *Arduino Electronics Blueprints*, Don Wilcher explains how a programmable logic controller (PLC) is used to operate various electronic and electromechanical devices that are wired to I/O wiring modules (<https://hub.packtpub.com/programmable-dc-motor-controller-lcd/>)

References and recommended reading

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